Ali Nikkhah

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Summary

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation. — Variant: Senior AI / Lead

Technical Skills

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
ML/Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper
Agentic: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE
LLM Serving/GPU: vLLM, TGI, AWQ, GPTQ, KV-cache
Data/MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray

Professional Experience

Iran Credit Scoring Bureau — Tehran, Iran
ML Engineer — Self-Hosted Graph RAG Assistant (Concurrent) — *Apr 2026 == Present*

- Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.
- Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.
- Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.
- Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.
- *Stack:* Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain, FAISS, vLLM
- *Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study*

Turquoise Digital — Tehran, Iran
Senior Data Scientist & Data Engineer (Concurrent) — *Sep 2025 == Mar 2026*

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, <60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, -22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — -78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- *Stack:* Python, PyTorch, Spark, Trino, PeerDB, Airflow, Great Expectations, Prometheus
- *Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV*

Advanced Analytics Australia — Remote · Australia (Tehran-based)
Computer Vision / ML Platform Engineer (Edge) (Concurrent) — *Sep 2025 == Apr 2026*

- Deployed YOLO + ViT quantized (TensorRT/TF Lite) on Triton — 99.2% uptime <50ms p99, 3.5x speedup, exactly-once Kafka, 30 FPS over 100+ streams.
- Built edge MLOps with Triton model repository, zero-downtime rolling updates, automated rollback, Prometheus/Grafana per-site health.
- Cut false positives -35% via attention-guided NMS + SNR monitoring; broker architecture for 12 distributed sites.
- *Stack:* Python, OpenCV, YOLO, ViT, TensorRT, TFLite, Triton, Kafka
- *Parallel remote contract alongside Turquoise — edge CV track*

Sharif University — Robust & Interpretable ML Lab — Tehran, Iran
Research Collaborator — Explainable & Robust AI (Concurrent) — *Jul 2025 == Sep 2025*

- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- *Stack:* PyTorch, MLflow, Captum, SHAP

– *Summer research sprint concurrent with industry roles*

Trinity College Dublin — EmoDub Project Dublin, Ireland
Machine Learning Researcher — Emotion-Aware Speech Translation (Concurrent) *May 2025 == Nov 2025*
– Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
– Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
– Collab. with Prof. Siobhán Clarke; reproducible codebase https://github.com/AliNikkhah2001/emodub-emotion-translation.
– *Stack:* PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper, EmotionNet, EmoDB
– *Part-time remote research concurrent with Digikala + Turquoise + UBC*

Digikala Tehran, Iran
AI Engineering R&D Lead (Concurrent) *Jan 2025 == Aug 2025*
– Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback.
– Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, -41% hallucination; multi-tier Redis caching, p99 <200ms.
– LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards.
– Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts.
– *Stack:* Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch, ChromaDB, HyDE
– *Full-time lead; UBC/Trinity/Sharif were part-time remote (<10h/wk)*

Digikala Tehran, Iran
AI Engineer (Concurrent) *Nov 2024 == Jan 2025*
– Implemented LangGraph orchestration layers with health monitoring; hybrid retrieval with RRF + deterministic fallbacks for auditability.
– Embedded eval telemetry, memory-efficient indexing, response caching; load-tested to 5M queries.
– *Stack:* LangChain, LlamaIndex, FAISS, Elasticsearch, FastAPI
– *Promotion track — converted to R&D Lead Jan 2025*

University of British Columbia Remote · Vancouver, Cana
Research Assistant — Vision-Language Medical Imaging (Ultrasound Report Generation) (Concurrent) *Feb*
– Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora.
– Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split.
– Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025).
– *Stack:* PyTorch, ViT, BLIP, T5, LLaVA, CLIP, OpenCV, MIMIC-CXR
– *Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise*

L3S Research Center — Leibniz University Hannover Hannover, Germany
Research Assistant — Video Motion Classification (Concurrent) *Apr 2023 == Feb 2024*
– Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift.
– Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy.
– *Stack:* PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models
– *Part-time RA during B.Sc. final year — no conflict*

HomaCloud Tehran, Iran
MLOps Engineer (Founding) (Concurrent) *Aug 2021 == Feb 2022*
– Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — <10 min releases, 99.9% deploy success, 15+ models.
– GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
– *Stack:* Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD, FastAPI, Prometheus
– *During B.Sc. — clean sequential after Mapna internship*

Selected Projects & Research

Trinity College Dublin · EmoDub May 2025 – Nov 2025
Emotion-aware voice translation

– SER front-end + wav2vec2.0 bullets (see experiences)

University of British Columbia Feb 2024 – Nov 2025
Ultrasound report generation

L3S Research Center Apr 2023 – Feb 2024
Texture-free motion intelligence

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning — *arXiv preprint — in preparation*, 2025

Education

Sharif University of Technology Tehran, Iran
M.Sc. Artificial Intelligence Computer Engineering (full-time, ongoing) *Sept 2025 – Present*

Sharif University of Technology Tehran, Iran
B.Sc. Electrical Engineering GPA: 3.65 *Sept 2020 – Sept 2025*